

ENTRY 079 Correcting the Kyiv/Brisbane “Regression” — September 16, 2026 — Status: COMPLETE

It Was a Split-Composition Artifact, Not a Real Capacity Effect

Entry 077's Reported Kyiv/Brisbane Dip Does Not Survive a Controlled Comparison

The Problem With Entry 077's Original Comparison

Entry 077 reported Kyiv MAE rising 0.77→1.08 and Brisbane 0.99→1.14 when Osaka joined training, read as the model trading a little same-chip precision for broader generalization. That comparison used two DIFFERENT random 80/20 KFold splits (seed=42 each time, but over datasets of different length — 5,690 graphs for Entry 069, 6,398 for Entry 077) — so the actual circuits landing in each "held-out test set" were not the same circuits. A difference in test-set composition alone can move per-chip MAE by this much; the comparison was confounded from the start.

Controlled Re-Test

Re-scored the Entry 077 combined model on the EXACT same 1,138 held-out circuits Entry 069 used (the first 5,690 rows of entry077_graph_dataset.json are in identical order to entry068_graph_dataset.json, confirmed by spot-check, so Entry 069's KFold test indices apply unchanged). This isolates the model-capacity question from split-composition noise.

Chip	Entry 069 (3-chip)	Entry 077 (4-chip), same test circuits	Delta
Kyiv	0.77	0.84	+0.07 (noise-level)
Sherbrooke	1.45	1.09	-0.36 (better)
Brisbane	0.99	0.80	-0.19 (better)
Overall	1.07	0.91	-0.16 (better)

MAE in points, on the identical held-out circuit set both times. Floor-collapse MAE on this same set: Kyiv=0.75, Sherbrooke=0.54, Brisbane=0.17 — all healthy, no regression there either.

Entry 079 — Conclusion

There is no real capacity-dilution effect here. When compared fairly — same circuits, same evaluation — the 4-chip model matches or beats the 3-chip model on every one of the original three chips, not just on generalization to the unseen fifth. Entry 077's "nuanced, not-uniform" framing was itself an artifact of an apples-to-oranges comparison. No architecture or weighting fix is warranted. The corrected, honest verdict: adding Osaka to training improved the model across the board — same-chip accuracy AND zero-shot generalization to a genuinely new chip — with no accuracy trade-off found on a controlled re-test.

Research Log — Index Addendum (Entry 079)

No new files — this entry is a controlled re-analysis of entry077_combined_params.json against Entry 069's original KFold test_idx, run directly against the existing dataset files.